

DISSERTATION PRACTICAL WORK EVALUATION METRICS

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Criterion	What you evaluate	Source
Functional Coverage	Does the architecture support all functional requirements in the SRS?	ISO/IEC 25010 (Functional Suitability)
Modularity	Are responsibilities separated into coherent modules with low coupling?	Bass et al.
Maintainability	Can the architecture be modified or extended easily?	ISO/IEC 25010
Scalability	Can the architecture handle increased workload?	Bass et al.
Security Consideration	Does the architecture include authentication, authorization, secure communication, etc., if required?	ISO/IEC 25010
Interoperability	Does it support interaction with external systems identified in the SRS?	ISO/IEC 25010
Requirement Traceability	Can each major requirement be mapped to architectural elements?	Software Architecture literature
Cross Case Analysis	To what extent can the requirements specified in the SRS be traced to architectural components within the proposed software architecture?	Overall Synthesis

Software Architecture Evaluation Rubric

1. Functional Suitability (ISO/IEC 25010)

Why this metric?

Functional suitability in ISO/IEC 25010 measures whether a software product provides the functions required by users. Since this dissertation evaluates **architectural designs rather than implemented software**, the metric is operationalised as **Functional Coverage**, which assesses whether the architecture contains sufficient components and interactions to support all functional requirements defined in the SRS.

Source

ISO/IEC 25010 (2011)

Evaluation Question:

Does the architecture support the functional requirements specified in the SRS?

Score	Observable Criteria
5	Appropriate architectural components and interactions represent all functional requirements. No significant functionality is missing.
4	Most functional requirements are supported. One or two minor functions are missing or only partially addressed.
3	Core functionality is supported, but several important requirements are not represented.
2	Many functional requirements are unsupported or poorly represented.
1	The architecture fails to support most functional requirements.

Evidence to collect

- Missing components
 - Missing business functions
 - Unsupported use cases
 - Requirement-to-component mapping
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2. Modularity (Bass, Clements & Kazman)

Why this metric?

Modularity is a fundamental architectural quality attribute. Bass, Clements, and Kazman explain that effective modularisation reduces complexity, improves maintainability, and enables independent development and evolution of software components.

Source

Bass, Clements & Kazman (2022)

Evaluation Question:

Does the architecture separate responsibilities into cohesive and loosely coupled modules?

Score	Observable Criteria
5	Clear separation of responsibilities; high cohesion; low coupling; each component has a single well-defined responsibility.
4	Good modularization with only minor overlap between components.
3	Moderate modularity; some components perform multiple unrelated responsibilities.
2	Poor modularity with considerable overlap and tight coupling.
1	No meaningful modular separation.

Evidence to collect

- Component responsibilities
 - Coupling between components
 - Separation of concerns
 - Component dependencies
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3. Maintainability (ISO/IEC 25010)

Evaluation Question:

Can the architecture be modified or extended with minimal impact on other components?

Score	Observable Criteria
5	Architecture is highly maintainable with independent modules and clear interfaces.
4	Mostly maintainable with minor dependencies between modules.
3	Acceptable maintainability but several modules require coordinated modification.
2	Low maintainability due to tightly coupled components.
1	Architecture would be difficult to modify or extend.

Evidence to collect

- Interface clarity
 - Component independence
 - Change impact
 - Coupling
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4. Scalability (Bass, Clements & Kazman)

Why this metric?

Scalability reflects the ability of an architecture to accommodate increased workload while maintaining acceptable performance. Bass et al. identify scalability as an important architectural quality influenced by component decomposition and deployment decisions.

Source

Bass, Clements & Kazman (2022)

Evaluation Question:

Can the architecture accommodate increased workload or future growth?

Score	Observable Criteria
5	Architecture clearly supports horizontal or vertical scaling with independently scalable components.

4	Most components can scale independently with minor limitations.
3	Some scalability support exists but important bottlenecks remain.
2	Limited scalability due to centralized or tightly coupled design.
1	Architecture provides little or no scalability support.

Evidence to collect

- Stateless services
- Load balancing
- Distributed processing
- Central bottlenecks

5. Interoperability (ISO/IEC 25010)

Why this metric?

Many enterprise systems rely on interaction with external services. ISO/IEC 25010 identifies interoperability as an important quality characteristic describing the ability of software systems to exchange information and operate with other systems.

Source

ISO/IEC 25010 (2011)

Evaluation Question:

Does the architecture support interaction with required external systems?

Score	Observable Criteria
5	All required external systems are integrated using clearly defined interfaces.
4	Most required integrations are represented.
3	Some external integrations are present, but important ones are missing.
2	Very limited integration support.
1	No meaningful support for external system interaction.

Evidence to collect

- External APIs
 - Payment gateways
 - Authentication systems
 - ERP/CRM integration
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6. Security Considerations (ISO/IEC 25010)

Why this metric?

Security is included only when security requirements are specified in the SRS. The architecture should demonstrate mechanisms that support confidentiality, integrity, authentication, and access control where appropriate.

Source

ISO/IEC 25010 (2011)

Evaluation Question:

Does the architecture include components or mechanisms to address the security requirements stated in the SRS?

Score	Observable Criteria
5	Security mechanisms fully address all stated security requirements (e.g., authentication, authorization, secure communication, audit logging).
4	Most required security mechanisms are present with only minor omissions.
3	Basic security mechanisms are included but several important aspects are absent.
2	Very limited security support.
1	No identifiable security mechanisms despite security requirements.

Evidence to collect

- Authentication
 - Authorization
 - Encryption
 - Audit logging
 - Secure communication
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7. Requirements-to-Architecture Traceability

Why this metric?

Requirements-to-Architecture Traceability evaluates whether the software architecture adequately represents and supports the functional and non-functional requirements defined in the Software Requirements Specification (SRS). A well-designed architecture should provide clear architectural elements that can be traced back to system requirements, ensuring that no significant requirements are omitted during architectural design.

Source

- **ISO/IEC/IEEE 29148:2018** – Systems and Software Engineering – Requirements Engineering
- **Bass, Clements & Kazman (2022)** – *Software Architecture in Practice* (4th Edition)
- **ISO/IEC/IEEE 42010:2022** – Architecture Description

Evaluation Question

To what extent can the requirements specified in the SRS be traced to architectural components within the proposed software architecture?

Score	Observable Criteria
5	All functional and relevant non-functional requirements can be clearly mapped to one or more architectural components. No significant omissions are identified.
4	Most requirements are represented by architectural components. Only minor requirements are partially represented or omitted.
3	Core requirements are represented, but several important requirements cannot be clearly traced to the architecture.
2	Many requirements cannot be mapped to architectural components, indicating incomplete architectural coverage.
1	Little or no traceability exists between the SRS and the software architecture.

Evidence to Collect

- Requirements-to-Architecture Traceability Matrix
- Functional Requirement → Component mapping
- Non-functional Requirement → Architectural decision mapping

- Missing architectural support for documented requirements
- Components with no identifiable requirement
- Requirements with no corresponding architectural element

8. Cross-Case Analysis

Why this analysis?

Cross-case analysis synthesises findings from multiple case studies to identify recurring patterns, similarities, and differences that cannot be observed from analysing a single case study alone. This approach increases the robustness of qualitative empirical research by identifying consistent trends across different contexts.

Evaluation Matrix

For each case study:

Criterion	Human Architecture	AI Architecture	Evidence
Functional Suitability			
Modularity			
Maintainability			
Scalability			
Interoperability			
Security			
Traceability			

Similarity Evaluation Matrix

Attribute	Human	AI	Similarity
Architecture Style	Microservices	Mircroservices	Same
Main Components	12	11	High
External Systems	5	4	Medium
Functional Coverage	Complete	Complete	High
Data Flow	Similar	Similar	High
Component Relationships	Different	-	Medium

Similarity Rubric

Score	Meaning
5	Nearly identical architecture with only minor differences
4	Similar architecture with some differences
3	Moderate similarity
2	Low similarity
1	Completely different architecture

Aspect	Similarity
Architecture Style	Different
Components	High
External Systems	Medium
Data Flow	High
Overall Similarity	Moderate

Example (Case Study 1)

Criterion	Human	AI	Evidence
Functional Suitability	5	4	AI omitted delivery notification component.
Modularity	5	4	AI combined payment and order processing.
Maintainability	4	4	Both architectures have well-defined interfaces.
Scalability	5	5	Both support independent service scaling.
Interoperability	5	4	AI omitted supplier integration.
Security	4	3	AI includes authentication but no authorization mechanism.
Requirements to traceability	5	4	Human Satisfies Requirements mentioned in SRS than
